

Declaration of ethical AI use for assignments

School of General Education- BRAC University

Weight- 5% of the Assignment Grade

Students are instructed to complete this form with sincerity, declaring that they have used AI in an ethical manner and as a learning tool to support the successful completion of this assignment. Completion of this form accounts for 5% of the total grade of this assignment. Failure to submit a declaration or an incomplete form will result in 0%.

AI is a tool to enhance and support your work, not to replace your own critical thinking, analysis, and creative effort. The final product must be a result of your/ your team's intellectual and creative labor. All AI use must be openly declared.

I hereby acknowledge that I have used the following Generative AI tools in the below described capacity to complete my assignment Final take home assessment for the CST301: For the Love of Food course which was due on 11th January,2026.

Part I: Student-declared AI awareness

Mention the AI tools that were used as support tools for this assignment (i.e., ChatGPT, Co-Pilot, Gamma, Gemini etc.)	ChatGPT, Claude, Gemini
Please write the initial ‘prompts’ you provided to the AI tool (i.e., “I am writing a book review for my course Emergence of Bangladesh. The book is titled X by Y. Please help me write a well-structured academic review of this book”)	1. Tell me about the pillar of food security. 2. How can a general pathway be created? 3. What is conditional commercialization? 4. Correct the grammar. 5. What is the question saying? 6. Suggest me better synonyms. 7. Rate my answer, tell me about the weak part. 8. Did I answer all the questions? 9. Give me the citation for this paper. 10. Give me a brief idea about GMO.

Part II: Instructor-approved options for AI use (Instructions for faculty: You may revise this list of options to fit your respective course/ assignment)

You are only allowed to use the aforementioned AI tools to **support** the following activities related to this assignment. Please choose carefully and **put a tick against the options** you have chosen to use AI for:

☐ Brainstorming Ideas

☐ Generating Outlines

- ☒ Editing Grammar and Style
- ☐ Summarizing Texts
- ☒ Formatting Citations
- ☒ Translating Texts

- ☒ Analyzing Data
- ☐ Generating Visual Aids
- ☐ Research Assistance (Finding secondary sources)

Part III: Instructor-approved AI restrictions *(Instructions for faculty: You may revise this list of options to fit your respective course/ assignment)*

You are **NOT** allowed to use AI for the following activities.

- | | |
|---|---|
| 1. Submitting AI-Generated Work as Original | 7. Violating Assignment Instructions |
| 2. Paraphrasing (Quillbot etc.) | 8. Plagiarizing with AI |
| 3. Falsifying Citations | 9. Automating Group Work |
| 4. Producing Unedited AI Content | 10. Creating Misleading Visuals |
| 5. Bypassing Critical Thinking | 11. Submitting AI-Generated Creative Work |
| 6. Generating False Data | |

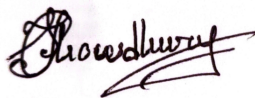
Part IV: Penalties *(Instructions for faculty: You may revise this list to fit your respective course/ assignment)*

Penalties for these violations include:

- Requirement to revise and resubmit work, potentially with a grade penalty.
- A failing grade on the assignment
- A reduced course grade.
- Referral to the Proctor's Office for disciplinary action

I understand the terms and conditions of this declaration and I hereby acknowledge that the information provided by me on this form is true and correct to the best of my knowledge

Samia Tabassum Chowdhury - 22301206



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Name, ID, & Signature of the student



Final-paper

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Section : 03

CST301 : For the Love of Food

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January 11, 2026

Q1. Using Sen's entitlement approach, argue whether famine is best explained by a failure of "food availability" or a failure of "food access." Your argument must link the claim to at least two pillars of food security and show one clear causal pathway from poverty to starvation.

Sen's entitlement approach revolutionized famine analysis by demonstrating that starvation results not from food absence but from the collapse of people's command over food. Sen argued that access to food operates through distinct entitlements: exchange entitlements (trading labor or goods for food), production entitlements (growing food directly), and transfer entitlements (receiving food through family or charity). The Bengal Famine 1943 illustrates breakdown of entitlement on disastrous levels. Despite a production decline to only 5 percent, the wage of agricultural workers to the rice exchange rate dropped by 100 to 34, which added to 3 million deaths (Islam, 2007). In 1974, a Bangladesh famine also made FAD logic more difficult: the availability of food grains had been 107 percent of that in 1971, but famine had occurred (Elahi, 2016). Nevertheless, the writer exposes methodological weaknesses in the data provided by Sen by cross-examining it. He states that Sen relied on 'guesstimated' data from agencies operating during the 1971 liberation war pandemonium, rendering the statistical base for comparing 1974 to 1971 fragmented and unreliable. Such data quality problems render invalid some of the empirical hypotheses, but the conceptual understanding is true, famine is inherently a question of access failure and not the problem of aggregate supply inadequacy.

Food security is the presence of sufficient food and access to it. The Bengal famine is a demonstration of the interactions between the availability and access pillars being disastrous in amplifying the modest scarcity. Nevertheless this small deficit caused a sense of uncertainty in the market which was ruthlessly taken advantage of by speculators. Prices of rice skyrocketed by a percentage from 11 to over 30 rupees per maund in months instead of increasing in line with the shortage (Islam, 2007). The Famine Inquiry Commission realized that the lack was not responsible for causing widespread starvation in normal conditions of distribution (Islam, 2007). Of crucial interest is the interaction of this small fall with the market forces and social organization to annihilate the people's control of food. This gap disproves the food shortage theory, showing a clear contradiction between a stable food supply and mass starvation that Sen's approach successfully explains. Conversely, the food access pillar demonstrates the actual mechanism of the catastrophic failure. The total collapse of agricultural laborers to exchange entitlement occurs due to the decrease in wage to rice ratio by 100 to 34 (Islam, 2007). Their power to sell labor to purchase food broke as the wages only rose by 25% and the food prices rose threefold. The collapse of the parallel entitlement of jute farmers was observed when the jute prices were not able to increase in the same proportion as the rice prices such that they were not able to buy sufficient food despite having marketable goods (Islam, 2007). Distress sales of land and livestock proved futile as asset values collapsed relative to skyrocketing rice prices, providing only temporary relief (Islam, 2007). Access failure was not confined to market mechanisms but social safety nets were also included. Charity networks and patron client

systems started to run on rails as the rural elite no longer felt compelled to provide sustenance to dependents. This demonstrates the inherent peculiarity of the Bengal crisis: rice was still on display in markets and millions of people were starving because they could not afford it, which proves that food security is not the amount of food produced but whether or not vulnerable populations can actually have access to it due to valid entitlements.

The causal pathway from poverty to starvation begins with initial vulnerability, followed by a macroeconomic shock, leading to the collapse of exchange entitlements, and culminating in physiological failure. Islam (2007) reveals 31% of rural Bengalis were badly nourished in normal times, consuming 400-460 grams of cereals daily with no reserves. The way jute farmers identified in blocked exchange paths because of falling behind in jute prices relative to rice inflation is recorded (Islam, 2007). Social safety nets failed to cope with the excessive demand, shutting even the non market access routes. The chronically poor had no entitlement left as wages were worthless, goods were sold and charity was out of resources. They had no physiological reserves that would help them to survive prolonged starvation, already being weakened by underlying malnutrition. The evidence indicates that the crisis was not a result of food scarcity, but rather an utter failure of the mechanisms required to access existing supplies. With sufficient aggregate supply, once exchange rates fall, when asset prices fall, when social channels are shattered, endangered populations are no longer able to assert the available food that continues to exist in the market, which confirms the observation by Sen that famine is a phenomenon of ruined entitlements to access rather than an absolute lack of food.

Q2. Using your conclusion from Q1, evaluate Bt brinjal (GMO) as a strategy to reduce hunger: does it reduce entitlement failure (by lowering costs/raising income), or does it create new vulnerabilities (biosafety, ecological risk, market dependency) that can worsen food security? Defend one side and rebut the strongest counterargument.

Based on the evidence provided in Bangladesh, Bt brinjal effectively reduces entitlement failure, as it promotes the economic entitlements of resource-poor farmers. The mechanisms that are involved in poverty-to-starvation transitions, as illustrated in the Bengal famine analysis, do not involve an ultimate food shortage, but systematic destruction of exchange entitlements.

Field evidence shows that Bt brinjal adoption has a significant entitlement enhancement. In a market value chain analysis conducted by Shelton et al. (2020) in five districts of Bangladesh, the analysis indicates that Bt brinjal yielded 19.6% higher yields and 21.7% higher revenue in yield and revenue per hectare, respectively, and equates to about 664 dollars in the form of additional income to resource-poor farmers. A BARI trial in 35 districts recorded even greater returns: Bt brinjal net returns of \$2,151 per hectare compared to non-Bt varieties of 357, which is sixfold higher (Shelton et al., 2019). This income boost directly counters the entitlement

collapse mechanism identified in Q1. The 1943 Great Bengal Famine demonstrated how a marginal 5% deficit in total rice supply (as estimated by the 1944 Famine Inquiry Commission) could trigger a catastrophic collapse of exchange entitlements when wage-rice ratios fell to roughly 34% of their pre-famine levels. Bt brinjal inverts this dynamic by raising farmer incomes 600%, it creates exchange entitlement buffers that can absorb price shocks. Resource-poor farmers gain command over food not through aggregate supply increases, but through enhanced purchasing power; precisely the access dimension Sen identified as critical. The dual mechanism of the technology deals with structural agricultural poverty in an all-inclusive manner. Bt brinjal lowers pesticide expenses by 61 percent and offers nearly 100 percent management of the eggplant fruit along with achieving nearly complete control of the eggplant fruit and shoot borer, whereas traditional insecticides fail despite farmers spraying up to 80 times per season (Shelton et al., 2019, 2018). Such cost cutting and productivity enhancement reinforces both production based entitlements to subsistence farmers and exchange entitlements to market dependent actors, which generates multi-layered vulnerability mitigation instead of single intervention solutions. Employment impacts also indicate value chain improvement in the form of entitlement. Increased productivity generated a 14% higher labor demand in harvesting and packaging, while superior fruit quality reduced the 'unsold' fraction from 8.7% to 3.6% (Shelton et al., 2020). These efficiency gains ensure production increases translate directly into income, addressing the vulnerability of agricultural laborers whose falling real wages triggered the 1943 famine.

The patterns of adoption can be considered to be one of the best confirmation of the effectiveness of Bt brinjal. An increase of 20 farmers in 2014 to 27,012 of total brinjal farmers in 2018 indicates a growth rate of about 17 percent of the total brinjal farmers in Bangladesh taking up in 4 years (Shelton et al., 2018). This voluntary adoption is critical since subsistence level farmers will not be able to pay off technologies that do not give quick results. According to the survey, 75.4% of existing Bt farmers and 71.4% of non-Bt farmers were planning to grow Bt brinjal, and that the satisfaction with Bt and non-Bt farmers with Bt was 83.1% yield and 80.6% fruit quality, respectively, which is far more than the satisfaction levels of yield among non-Bt farmers (58.7), which was low (Shelton et al., 2020). These uncovered preferences indicate that technology increases food security by the means that farmers understand and appreciate. The ex-ante Indian studies revealed 77 percent willingness to adopt by the farmers, of which 93 percent of them were already knowledgeable about the benefits of Bt cotton, but India has a moratorium on the study, which hinders India in realizing these potential benefits (Sahoo et al., 2021).

The strongest counterargument claims Bt brinjal creates three new vulnerabilities that could worsen food security: biosafety risks, ecological disruption, and market dependency (Bandopadhyay et al., 2012; Sahoo et al., 2021). Critics document 15% caloric reduction in Bt brinjal, rat studies showing diarrheal effects, and Cry protein allergen sequences. In South Asia's center of origin for brinjal, genetic contamination risks disrupting

biodiversity through wild-type introgression (Sahoo et al., 2021). Market dependency concerns center on corporate seed control, potentially trapping farmers in Mahyco's proprietary system. However, three factors demonstrate these concerns do not outweigh entitlement benefits. First, BARI's seven-year pre-commercialization testing across multiple locations revealed no ecological damage to non-target arthropods, while post-commercialization monitoring confirms statistically similar biodiversity in Bt and non-Bt fields (Shelton et al., 2019). Bt brinjal's 80% pesticide reduction represents measurable harm reduction, not hypothetical risk introduction. Third, the seed dependency argument collapses under Bangladesh's implementation model. Unlike hybrid crops, the four approved Bt varieties are open-pollinated lines enabling seed saving, distributed free by BARI/DAE or at <\$0.10/gram through BADC (Shelton et al., 2020). This public sector model contrasts sharply with India's corporate hybrid approach and eliminates the Mahyco monopoly concern. Farmers face greater dependency on expensive pesticides under conventional farming than on affordable/free Bt seeds.

Bt brinjal demonstrably reduces hunger vulnerability by strengthening resource-poor farmers' economic entitlements. While biosafety monitoring must continue, the technology addresses documented, immediate harms while delivering tangible benefits that farmers themselves validate through adoption and satisfaction.

Q3. Based on your evaluations from Q1 and Q2, take a position on this debate: “Conditional commercialization of GMO crops improves food security more than a temporary ban.” Justify your stance by weighing poverty-driven access risks against biosafety/governance risks and conclude with a clear yes/no decision.

Evidence demonstrates that conditional commercialization of GMO crops improves food security more effectively than temporary bans. As the Bt brinjal case in Bangladesh indicates, when properly governed conditional commercialization is more effective than needed precautionary bans in the strategy to guarantee food security as long as there is access and safety by the governance systems. Following the efforts of Q1 to reveal the failure of access and not availability as the cause of famine and Q2, the evidence that Bt brinjal improves the economic entitlements, this analysis balances the risks of poverty-based access against biosafety and governance issues.

As we discussed, Sen's framework proves famines result from exchange entitlement failure rather than supply shortages. This can be seen in the Bengal famine: even though production fell by only 5% agricultural workers' wage to rice ratio fell to 34 percent leading to mass starvation (Islam, 2007). This trend is supported by the 1974 Bangladesh famine, as it happened even though the amount of food in the aggregate supply was 107 percent of the levels in 1971, since the flood damages in the region and higher prices than wages wiped out access (Elahi,

2016). These historical instances put in place a vital lesson: food security relies not only on the economic accessibility of vulnerable populations to food at hand but on aggregate supply. Bt brinjal directly works on this access dimension by enhancing the entitlements of exchange. The technology provides 6 times fold net returns and 61 percent reductions in pesticide costs, 27,012 farmers have adopted the technology in four years, as established in Q2 (Shelton et al., 2018, 2020). Such economic benefits make them less vulnerable to the exact entitlement failures which caused famines in the past. Timely prohibitions would deprive these desperate farmers of such essential advantages, and would keep their susceptibilities, which bring about famines, alive. Modern resource-deprived farmers are the same in terms of precarity. Prohibitions only guarantee the maintenance of status quo of pesticide aggressive use, loss of crops, and income insecurity at the time that farmers require alternatives the most. It is not only the lost opportunity but proactive, continuous breeding of vulnerability.

Biosafety concerns raised by critics warrant serious consideration. Bandopadhyay et al. (2012) documented 15% caloric reduction in Bt brinjal, plus rat studies showing diarrheal effects and liver alterations, while Cry proteins contain amino acid sequences matching known allergens. Conditional frameworks enable systematic risk management through phased implementation and continuous monitoring. BARI's seven-year pre-approval testing across multiple geographic locations, followed by post-commercialization monitoring revealing statistically similar non-target arthropod densities in Bt and non-Bt fields, demonstrates how conditional approval addresses ecological concerns through evidence rather than precaution (Shelton et al., 2019, 2018). Crucially, baseline comparison matters: traditional brinjal farming requires 80 insecticide applications per season, creating documented health risks through pesticide exposure and residues. Theoretical Cry protein concerns must be balanced against real, quantifiable harms from intensive chemical use (Shelton et al., 2018).

Conditional frameworks enable adaptive management through refuge planting, baseline susceptibility monitoring, and Bt gene pyramiding (Shelton et al., 2019). Despite procedural inefficiencies (Shelton et al., 2018), conditional approval enables accountability and 'learning through implementation,' providing vital data-driven benefits that blanket bans eliminate. The strongest case for temporary bans involves governance capacity. Elahi (2016) demonstrates Bangladesh's weak, disintegrated statistical system at independence, where inconsistent data collection compromises policymaking. When biosafety surveillance depends on such systems, oversight validity becomes questionable. However, the argument contains a paradox, it advocates abandoning useful technology rather than building stronger governance. Weak institutional capacity argues for developing better institutions, not denying farmers yield-enhancing, pesticide-reducing technology while maintaining poor governance. India's indefinite Bt brinjal moratorium since 2010 exemplifies this trap. Indian farmers continue intensive pesticide applications while Bangladeshi farmers benefit from reduced chemical use and increased

incomes. This comparison reveals that bans do not halt harm; they sustain it by eliminating learning opportunities.

The main problem involves striking a balance between the access risks due to poverty and the biosafety risks and governance risks. The risk of access is timely, documented and fatal. These are actual, measurable damages, and not speculative dangers in the future. Although biosafety risks are a concern that needs to be observed, they are rather speculative a few years after enforcement. Hypothetical environmental damages have not been proven by post commercialization surveillance. The problems of governance are real but can be addressed by strengthening the institutions and not by denying technologies.

The fundamental imbalance lies between immediate, fatal poverty-driven access risks and speculative biosafety concerns that are manageable through monitoring. Deterrence of the current evils is better than uncertain futures. The facts are very clear that conditional commercialization is better than temporary bans. Bt brinjal prevents famine by strengthening economic entitlements, whereas bans sustain harm by denying farmers these benefits while allowing toxic pesticide practices to persist. The voluntary adoption by thousands confirms the technology's value, supporting the conclusion that yes, competent commercialization is more effective than precautionary bans.

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